

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

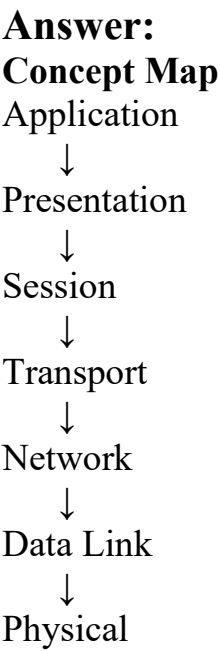
6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation



Missing Layers

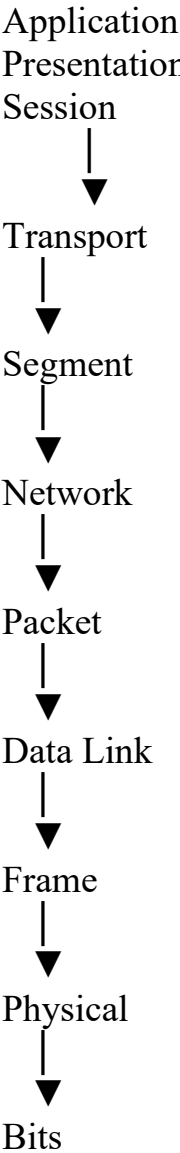
- Session Layer
- Data Link Layer

Analysis

- **Session Layer:** Establishes, manages, and terminates communication sessions. It also provides synchronization and recovery.
- **Data Link Layer:** Converts packets into frames, performs error detection, and controls access to the physical medium.
- Together, these layers help provide reliable and organized communication between devices.

2. Data Units and OSI Layers

Concept Map



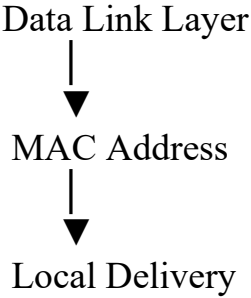
Analysis

- **Transport Layer** → Segment: Divides data into segments.
- **Network Layer** → Packet: Adds IP address for routing.
- **Data Link Layer** → Frame: Adds MAC address and error detection.
- **Physical Layer** → Bits: Converts frames into binary signals for transmission.

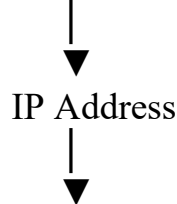
Each transformation prepares the data for the next stage of communication.

3. MAC Address and IP Address

Concept Map



Network Layer



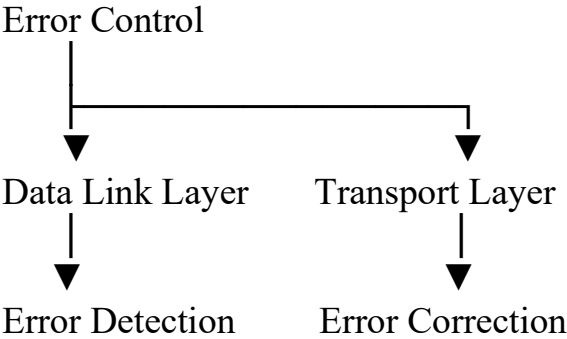
Global Delivery

Analysis

- **MAC Address:** Identifies devices within the same local network (LAN).
- **IP Address:** Identifies devices across different networks (Internet).
- Together, MAC and IP addressing ensure data reaches the correct destination both locally and globally.

4. Error Control

Concept Map



Analysis

- **Data Link Layer:** Detects errors using CRC or checksum.
- **Transport Layer:** Corrects errors by retransmitting lost or damaged data.
- These layers work together to improve reliable data transmission.

5. Web Browsing Using OSI Layers

Concept Map

Application

(Web Browser - HTTP/HTTPS)



Presentation

(Encryption - SSL/TLS)



Session

(Session Management)



Transport

(TCP)



Network

(IP Routing)



Data Link

(Frames & MAC Address)



Physical

(Cables/Wi-Fi Signals)

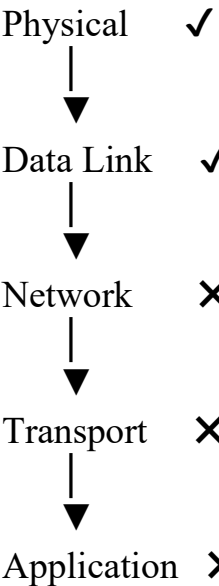
Analysis

- **Application Failure:** Website cannot open.
- **Presentation Failure:** Secure connection (HTTPS) fails.
- **Session Failure:** Connection is interrupted.
- **Transport Failure:** Data loss or incomplete webpage.
- **Network Failure:** Packets cannot reach the destination.
- **Data Link Failure:** Frames are corrupted.
- **Physical Failure:** No network connection.

Failure in any layer can interrupt the entire communication process.

6. Network Troubleshooting

Concept Map



Analysis

- **Physical Layer ✓:** Cable/Wi-Fi is working.
- **Data Link Layer ✓:** Frames are successfully transmitted.
- **Network Layer ✗:** IP routing has failed.
- Since the Network Layer fails, the Transport and Application layers also fail because data cannot reach its destination.
- The concept map helps identify that the problem starts at the Network Layer, making troubleshooting faster and easier.